Tobias C. 6/5/26

Image Processing Findings

When doing the lab, the surprising facts that I found while doing it is the fact that digital images can be compromised of a large collection of numerical values as opposed to a visual object. Color images tend to store red, green, and blue values for each pixel, while in grayscale, it depends on the intensity value per pixel. This can help determine the image dimensions, data type, and the individual pixel values found in these photographs. Mathematical operations can produce these visible effects by modifying their values. This is because these operations can be able to change the brightness and the contrast of the picture depending on the factors modified. Convulsion also used a series of complex operations since each output pixel depended on the neighboring pixels and the values in the kernel. Speaking of kernels, blue kernels made an average based on the nearby information, sharpening made differences while edge-detection saw rapid changes within intensity. A very complex concept that I learn ed from the lab was the utilization of the histogram equalization. This is because it does the operations on the basis of the average overall distribution of the pixel's vales instead of an individual object that is featured in the said image, and when comparing original and enhance version of the histograms created a clear picture on how the spread of the intense values based on mathematical operations can affect the contrast of the image.

The lab that I did can be connected to the Nano Banana demonstration, and this is because traditional image processing within machine learning-based AI systems can give these systems the capability to identify what is shown in the image via the utilization of numerical values and mathematical operations while combining traditional methods, which include the utilization of filtering, edge detection, and color adjustment among others. This gives machine learning systems the capability to identify complex patterns and relations based on the data obtained. This can have real world applications, and some of the real-world applications that are use these techniques include object recognition, medical image analysis, manufacturing inspection, autonomous vehicles, and document scanning among other things. It can also be used to improve the quality of images if they were taken in a dark environment as well as a way to correct camera angles. When using these techniques in a future project, I could have the capability to combine traditional methods with machine learning computing systems to upscale and improve the quality of raw images while using tradition processing as the starting point and then machine learning systems to perform image refinement for better quality images.

In conclusion, the way that machine learning can do image enhancement is what interests me the most, and this is because that through mathematical and numerical operations, the system can be able to create significant changes to the quality of the image. Before the lab, I thought that images can be edited through programs like GIMP and Photoshop, but with machine learning, the use of mathematical operations reduced the time and tasks it takes to refine an image, and that digital photography depends on the pixel values, color channels, or spatial coordinates, all of which can be affected by machine learning. Finally, what I want to know more from this lab is how machine learning algorithms can used all of this to do object detection and style transfer powered by neural networks, and how it's interesting to see machine learning algorithm systems preserve crucial information during the image enhancement.